

Computational Molecular Biology Methods

1.1 Homology Modeling

This chapter offers an overview on the computational methods used in the course.

1.1.1 Principle of homology modeling: advancements and limitations

Homology or comparative modeling methods are based on the idea that evolutionary related proteins share a similar structure (Chothia and Lesk, 1986).

The quality of the predicted models is thus correlated with the evolutionary distance between the template (member of the family with known structure) and the target proteins (Tramontano et al., 2008).

Homology model techniques are the most reliable and accurate methods to generate structures (Meier and Söding, 2015 and Tramontano et al., 2008). Broadly speaking, comparative modeling can be considered as a cost-effective alternative tool when experimental structures are absent. Although homology modeling can boast a number of successes in many applications, it should be noted that the generated models are predictions and could present some inaccuracies.

In this subchapter we discuss the main advantages and limitations of the comparative modelling techniques (Busato and Giorgetti, 2016). It is well-known that homology modeling methods depend strongly on both the sequence identity (SI) between the target and the templates and the accuracy of alignment. SI < 50% generally leads to structural divergence between the models and the actual experimental structure, measured as C α atom RMSD, larger than 1 Å (Chothia and Lesk, 1986). Actually, two proteins with SI > 35% were shown to share the same fold (Orengo et al., 1997). Finally, with low target-template SI (between 10% and 25%), the comparative models might contain serious errors, thus, it is strongly recommended the introduction of experimental information such as ligand information, site-directed mutagenesis, and other experimental restraints to improve the accuracy of the model (Yarnitzky et al. 2010). These very general observations are often sustainable but can vary depending on the protein of interest. Indeed, it is well-established that some “superfolds” dominate the fold space. This implies that even very distant proteins, e.g. GPCR proteins, can share a similar 3D structure (Magner et al., 2015). The limitations/errors in homology modeling can be grouped into five categories (Fiser, 2010 and Palomba and Cavasotto, 2015): (1) Errors in side-chains modeling. These errors can introduce drastic changes in the side-chains involved in the ligand binding (Rodrigues et al., 2013). The latter highlights the importance of using existing ligand information in the homology modelling protocol (Yarnitzky et al., 2010). (2) Structural Deviations of a target region that is aligned correctly with the template. Even if the aligned segments of the model are correct; the target protein could present local structural differences than the template structure, indeed these divergences could be due to artifacts in structure determination of the template in different environments, and not to errors in the alignment. Therefore, models might be improved by using unrestrained MD simulation in order to arrange the mis-folded regions (Schlegel et al., 2005) or by using multiple-template approaches (Srinivasan and Blundell, 1993). (3) Inaccuracies of

target regions that do not have an equivalent segment in the template structures. Indels are an important issue in the homology modeling procedure. For instance, residues in incorrect positions or missing residue in the comparative models could be due to gapped residues in the core region. (4) Distortions or shifts of a region that is aligned incorrectly with the template structures. The quality of the alignment is one the major problems in homology modeling, especially when the SI falls below 20%. Misaligned regions correspond to errors in positioning the target residues on the template fold; resulting in an unreliable model. Multiple sequence alignment (MSA) and hidden Markov models (HMM) profiles approaches combined with manual inspection and curation of the alignments are strongly recommended to investigate possible errors and adjust key motifs in the alignment. (5) Templates are fragmentary or incorrect. The misfolded structure resulting from using an erroneous template or fragmentary template is the problem typically arises when SI is below 30%. Moreover, for large protein (>200 residues) the number of full-length templates is still sparse comparing with the many fragments of full-length proteins”.

1.1.2 Steps in model generation

It is commonly accepted that the comparative modeling procedure involves the following steps (Fig. 3.1): (1) identification of proteins evolutionarily related to the target sequence, whose 3D structures are solved (templates); (2) target-template sequence alignment; (3) generation of an initial model for the target based on the template selected and the sequence alignment; (4) refinement of the model at the side chain, loops and backbone level by using MD simulations and incorporation of additional experimental structural information; (5) validation and evaluation of the model. These steps must be iterated to achieve an acceptable model.

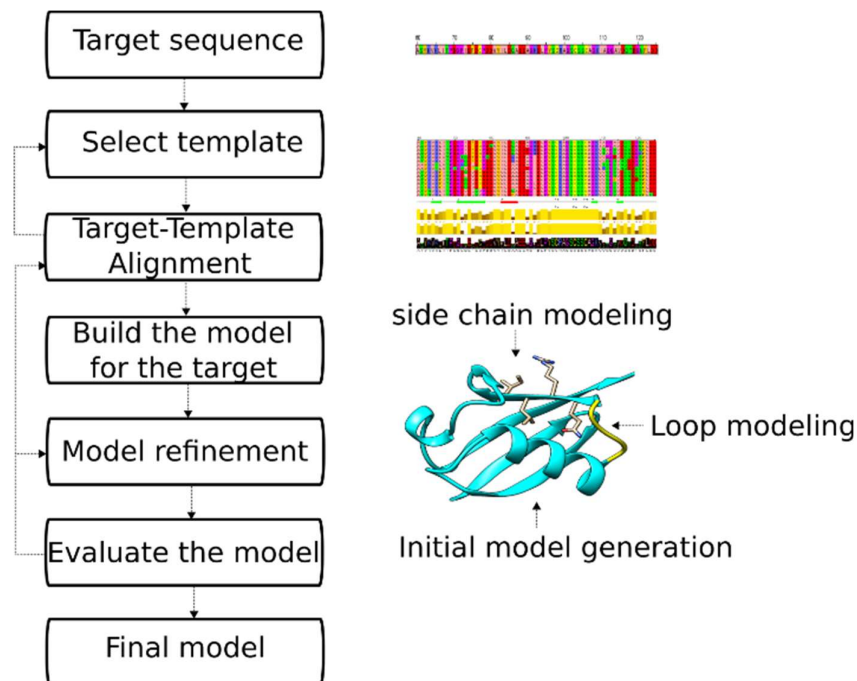


Figure 1.1. Workflow of the basic homology modeling procedure. Adapted from Busato and Giorgetti, 2016.

1.1.2.1 Template search and target-template sequence alignment

In homology modeling, template search and target-template sequence alignment should be considered as two independent and separate stages of the process. Actually, many homology modeling programs tends to integrate these two step. This because the identification of a 3D structure used as template is based on sequence alignment aided by database search techniques. As indicated in the Eswar *et al.*, 2008 paper, the sequence-structure relationship can be grouped into three different categories in the sequence similarity spectrum. The first group comprises the “easily detected relationships”, characterized by SI higher than 30%. As shown in Sander and Schneider, 1991 paper, most protein pairs (90%) sharing more than 30% SI were found to have similar structure. The second group is the so-called “twilight zone” (Rost, 1999), compatible to the 10% to 30% range of SI. When the SI falls in the twilight zone, the statistical measure for the evolutionary relatedness of proteins can be unsure (Rost, 1999). The third group represents the “midnight zone” (Rost, 1999), corresponding to statistically irrelevant sequence similarity. Depending on different regimes, different sequence alignment methods can be used. Historically, the alignment methods can be divided in three group:

1) *Pairwise sequence alignment methods.* Pairwise sequence alignment methods search for the global or the best local alignments of two sequences (Eswar *et al.*, 2007).

While, the global alignments try to align the entire length of sequences, the local alignments only search highly similar portions within long sequences. By definition, pairwise alignments can only align two query sequences at a time, and they can be exploited for searching in a database the sequences with high similarity to a query sequence. The two most famous pairwise sequence alignment methods aided by database search techniques are FASTA (Lipman and Pearson, 1985) and BLAST (Altschul *et al.*, 1997).

2) *Profile-sequence alignment methods.* With the profile-sequence alignment algorithms the query sequence is compared with a sequence profile, which typically represents a protein family and contains the evolutionarily relevant features of the family. The sequence profile is calculated from a MSA and it is represented as a position-specific scoring matrix (PSSM) (Henikoff and Henikoff, 1994 and Altschul *et al.*, 1997) or as a HMM (Krogh *et al.*, 1994 and Eddy, 1998). In this approach, instead of using only a single sequence, a frequency profile is used to filter the sequence information that is not conserved in the protein family. These algorithms are more efficient than pairwise sequence alignment methods in finding all distant homologues. They can recover roughly twice the number of homologs under 40% of SI (Eswar *et al.*, 2007). The BLAST variant i.e. PSI-BLAST use this approach (Altschul *et al.*, 1997).

3) *Profile-profile alignment methods.* These methods work in two steps: i) they generate a sequence profile of the target and then they compare it to the sequence profiles of solved structures. As stated in the Remmert *et al.*, 2012 paper, the profile-profile alignment algorithms are the most efficient approach for choosing and aligning templates in the

homology modeling process. This fact was confirmed by Prof. Marti-Renom, who pointed out that “Profile-profile methods detect about 28% more relationships at the super family level and improve the alignment accuracy for 15% to 20%, compared to profile-sequence methods” (Marti-Renom *et al.*, 2004; Zhou and Zhou, 2005 and Söding, 2005). These algorithms are implemented in the main server and software for protein structure prediction. Indeed, HHpred, the first server that is based on the pairwise comparison of profile HMMs, turned out to be the best-scoring server in template-based structure prediction (Mariani *et al.*, 2011) in CASP9.

1.1.2.2 Template Selection

Once a list of suitable structural homologues templates (retrieved from PDB database) and their alignments with target have been realized, at least five strategies exist to select the best template(s) (Tramontano *et al.*, 2006). The main strategies are:

- 1) select the template showing highest sequence similarity with the target.
- 2) build a “theoretical template” by using the average of the coordinates of all possible templates.
- 3) take the conformation of different regions from a given set of templates, in such a way that the local similarity is highest in each region;
- 4) create a structural prediction from each template and select the best model according to some criteria (see 3.1.2.4 Model Quality Assessment Section)
- 5) derive constraints from the templates and then create a structural prediction that satisfies these spatial restraints.

1.1.2.3 Model construction and refinement

Once a template-target alignment is made, the coordinates of the target can be built. There are three methods to generate the initial structural model: i) Rigid body assembly; ii) Segment matching; iii) Satisfaction of spatial restraints.

Some popular computer programs for homology modeling (Martí-Renom *et al.*, 2000) are based on rigid-body assembly method. In this strategy, a homology model of query portions is built up from a set of rigid bodies determined from the aligned 3D template. The most software based on this strategy are SwissModel (Biasini *et al.*, 2014; Arnold *et al.*, 2006; Kiefer *et al.*, 2009 and Guex *et al.*, 2009), RosettaCM (Song *et al.*, 2013), WHAT IF (Krieger *et al.*, 2009 and Vriend, 1990), Prime (<http://www.schrodinger.com/Prime/>), COMPOSER (Sutcliffe *et al.*, 1987), CONGEN (Brucoleri, 1993), ICM (<http://www.molsoft.com/>), PrISM (Yang and Honig, 1999) and 3D-JIGSAW (Bates *et al.*, 2001).

SegMod (Levitt, 1992) is the most notorious software based on segment matching approach. The latter builds the crude model by using a subset of atomic positions e.g. C α from the 3D template (Palomba and Cavasotto, 2015).

Finally, the most famous program, which generates models by satisfaction of spatial restraints is MODELLER (Fiser et al., 2000; Martí-Renom et al., 2000; Šali and Blundell, 1994 and Webb and Sali, 2014). MODELLER was used in this thesis, so we decided to provide a short description. The algorithm uses a set of spatial constraints such as interatomic distances, ϕ and ψ and backbone angles recovered from the experimentally solved structure and subsequently applies them to the target sequence, each with an associated probability density function (pdf 's) (see Fig.3.2). These structurally-derived constraints are so combined with general spatial constraints, derived from a CHARMM force field, into an objective function. Therefore, the model is built minimizing the violations of the geometric restraints. In MODELLER, the minimization step is carried out in two phases. The first uses the variable target function method. Then, the structural prediction is refined by MD with simulated annealing (SA).

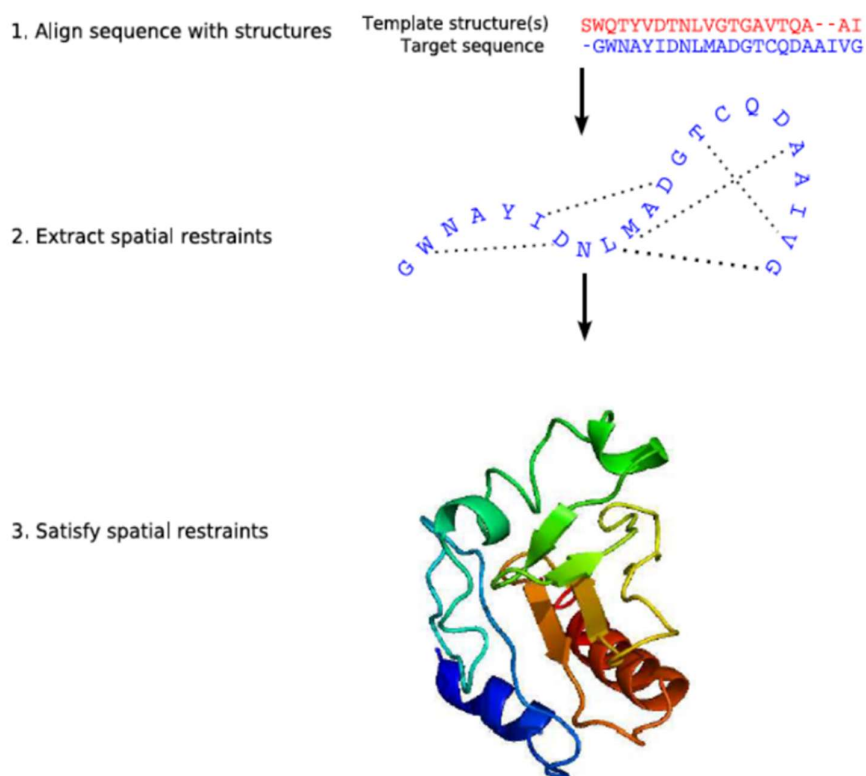


Figure 3.2. Basic steps of the MODELLER procedure. Adapted from the MODELLER 9v12 manual (<https://salilab.org/modeller/9.15/manual.pdf>).

1.1.2.4 Model quality assessment

The 3D structural predictions may contain errors, thus some features must be checked with attention in the model. The main model quality assessment factors/tools are briefly presented as follows:

- *Sequence identity.* The percentage of SI between target and template is a good predictor of the accuracy of a model. Model accuracy steadily increases with increasing SI.

- *Stereochemistry check.* The most common programs that evaluate the stereochemistry (e.g., bond-lengths, bond-angles, backbone torsion angles, and non-bonded contacts) of the structural prediction are PROCHECK (Laskowski *et al.*, 1993) and WHATCHECK (Hooft *et al.*, 1996). As said by Prof. Fiser, even though errors in stereochemistry are usually unusual and less informative than errors identified by statistical potentials, a cluster of stereo chemical errors might indicate that there are larger errors (e.g., alignment errors) in that portion (Fiser, 2010).
- *Global model quality estimation.* Global indicators of the quality provide a pseudo-energy for the structural prediction. The most common global indicators are: 1) DFIRE (Zhou and Zhou, 2005), an all-atom distance-dependent statistical potential; 2) QMEAN (Benkert *et al.*, 2009) a composite scoring function for model quality estimation and 3) DOPE (Shen and Sali, 2006), a statistical scoring function for model assessment based on an atomic distance-dependent statistical potential derived from ca. 1500 crystallographic structures.
- *Local model quality estimation.* Here, the accuracy of specific segments of the model is checked, identifying potentially incorrect portions in the structural prediction. The most famous tools are: 1) ProQres (Wallner and Elofsson, 2006), an artificial neural network trained to guess the local 3D prediction quality based on some structural characteristics; 2) ANOLEA (Melo and Feytmans, 1998), a statistical potential that can be exploited to investigate the packing quality of the model and 3) GROMOS (Christen *et al.*, 2005), an empirical force field that is mainly used to calculate the energy of each residue in the model.

1.2 Molecular Docking

1.2.1 Basic concepts of molecular docking

Proteins do not occur as isolated but are exposed to many interactions. In order to predict the biological function/activity of a protein, a detailed study of its interactions with its partners (proteins or small organic molecule) is required. The computational prediction of multimolecular complex structures is called “*docking*”. The binding pose of a ligand against the receptor can be uniquely described by its state variables: a number of parameters that fully describe the ligand position. The state variables of a ligand represent its coordinates, orientations, flexibility and structural conformations. A state variable represents one degree of freedom in a multidimensional search space, and their bounds describe the extent of the search.

Historically, simplified algorithms were developed to reduce the docking process, by keeping rigid the molecules along the docking protocol. However, it is important to highlight that these methodologies might produce not-reliable results (poses) when the starting conformation of the ligands is wrong.

Broadly speaking, all the docking algorithms share two key components: i) a scoring function, which is maximized (minimized) to rank the complexes and ii) a search method to explore the state variables. In principle, the docking technique consists in a search of the conformational space the most energetically favorable pose of the ligand. In practice, this would mean a full exploration of the components' energy landscape, resulting in an

unrealistic approach due to the high computational time and cost. A classic example is the “virtual screening”, a methodology that requires the calculation of protein binding poses of many ligands. Empirical, force field-based or knowledge-based scoring functions have been developed to reduce the size of the conformational space. These scoring functions can be divided into two groups: i) systematic and ii) stochastic.

Prof. Garrett M Morris in the “docking and virtual screening” book (Morris et. al., 1998) clearly defined these two terms as follows: while systematic search methods sample the search space at predefined intervals and are deterministic, the stochastic search methods iteratively make random changes to the state variables until a user-defined termination criterion is met.

The search methods can be further divided in “local” and “global” depending on how extensively they can explore the search space. While the local search methods attempts to detect the nearest or local minimum energy to the current conformation, the global search methods look for the global minimum energy within the defined search space (Morris *et al.*, 1998).

1.2.2 Information-driven docking with HADDOCK

The docking procedure is based on the use of constraints resulting from experimental information in order to predetermine the surface of interaction between the macromolecules in question. HADDOCK (Van Zundert et.al, 2016 and Dominguez et.al., 2003) can be considered the most successful software that uses an information-driven approach, written and maintained by Alexandre Bonvin and coworkers. Originally conceived for protein-protein docking; now HADDOCK (Van Zundert et.al., 2016 and Dominguez et.al., 2003) has been applied at small molecule/protein docking. However, when small organic molecules are included in the docking phase, the force field topologies and parameters for these molecules must generated before to run a docking experiment. Two state-of-the-art software i.e. PRODRG (SchuÈttelkopf and Van Aalten, 2004 and Van Aalten et.al., 1996) and ACPYPE (da Silva and Vranken, 2012) can perform this task.

While PRODRG (SchuÈttelkopf and Van Aalten, 2004 and Van Aalten et.al.,1996) is based on a database search method to parametrize the molecules using OPLS-like parameters (Paulechka et.al, 2012), ACPYPE (da Silva and Vranken, 2012) uses ANTECHAMBER (Wang et.al., 2006) with a semi-empirical quantum calculation method for the partial charges.

Ambiguous Interaction Restraints (AIRs). The experimental data (obtained by site-directed mutagenesis, NMR, bioinformatics predictions, etc.) are introduced in the form of ambiguous constraints of interaction (AIRs, Ambiguous Interaction Restraints), defined as ambiguous distances between all the residues involved in the interaction. An AIR causes a residue on the surface of a biomolecule that is lead in the vicinity of another residue or group of residues on the molecule partners, when they form the complex. An AIR is defined as an ambiguous intermolecular distance (d_{iAB}) with a maximum value of 3 Å between any atom m of an active residue i of molecule A (m_{iA}) and any atom n of both active and passive residues k (N_{res} in total) of molecule B (n_{kB}) (and inversely for molecule B). The effective distance d_{iAB}^{eff} for each restraint is calculated using the equation:

$$d_{iAB}^{eff} = \left(\sum_{m_{iA}=1}^{N_{atoms}} \sum_{k=1}^{N_{res}} \sum_{n_{kB}=1}^{N_{atoms}} \frac{1}{d_{m_{iA}n_{kB}}^6} \right)^{(-1/6)} \quad 8$$

Equation 1.1. Effective distance (d_{AB}^{eff}) for each restraint in HADDOCK

The N_{atoms} indicates all atoms of a given residue and N_{res} the sum of active and passive residues for a given molecule. In this way, the passive residues do not have direct AIRs to the partner molecule but can satisfy the partner molecule active restraints. A $1/r_6$ sum averaging is used because this mimics the attractive part of a Lennard-Jones potential and ensures that the AIRs are satisfied as soon as any two atoms of the two molecules are in contact. The 3Å limit represents a compromise between hydrogen-hydrogen and heavy atom-heavy atom minimum van der Waals distances. The use of AIR allows HADDOCK (Van Zundert et.al., 2016 and Dominguez et.al., 2003) to search through all the possible configurations around the interacting site defined by the information that we have about the problem in question. The AIRs are incorporated as a term of additional energy to the energy function (scoring function) that is minimized during the docking. It will still set an upper limit of distance, typically 2 Å, below which the constraint is satisfied and the attractive force is zero; if the limit is exceeded, the pairs of atoms are subjected to an attractive force that leads them to approach. Because some interatomic distances inversely contribute to the actual distance, an AIR is typically satisfied if a pair of atoms, part of a constraint, is between 3 and 5 Å, depending on the degree of ambiguity. In the definition of those residues, one can distinguish between “active” and “passive” residues. The “active” residues are essential for the interaction, based on experimental data. The “passive” residues are all solvent accessible surface, neighbours of active residue. These residues must be identified and specified by the operator. If experimental data are missing, bioinformatics is needed in the definition of active and passive residues. The most important bioinformatics tool for prediction of protein-protein interface residues is CPORT (de Vries and Bonvin, 2011). The latter combines six interface prediction methods WHISCY (de Vries et. al., 2006), PIER (Kufareva et.al., 2007), ProMate (Neuvirth et.al.,2004), cons-PPISP (Chen and Zhou,2005), SPPIDER (Porollo and Meller, 2007), and PINUP (Liang et.al., 2006) into a consensus predictor. CPORT (de Vries and Bonvin, 2011) results can be exploited as active and passive residues in HADDOCK (Van Zundert et.al., 2016 and Dominguez et.al., 2003).

Docking Protocol. The program is limited to optimize the interaction between the two molecules in the space of the constraints provided by the operator. The protocol of docking, which requires the pdb files of the free molecules and AIRs, comprises three steps:

1. randomization of orientations and rigid body energy minimization (EM);
2. semi rigid simulated annealing in torsion angle space (TAD-SA);
3. final refinement in Cartesian space with explicit solvent.

The first docking step in HADDOCK (Van Zundert et al., 2016 and Dominguez et.al., 2003) is a rigid body energy minimization. First the molecules will be separated by a minimum of 25Å and rotated randomly around their centre of mass.

The rigid body minimization is performed stepwise: 1) four cycles of rotational minimization in which each molecule (molecule and associated solvent in case of solvated

docking) is allowed to rotate in turn; 2) two cycles of rotational and translational rigid body minimization in which each molecule and associated solvent is treated as one rigid body.

The second docking step is semi-flexible SA. The best 200 structures after rigid body docking will be subjected to a semi-flexible SA in a torsion angle space. The semi-flexible annealing consists of several stages: 1) High temperature rigid body search; 2) rigid body SA; 3) semi-flexible SA with flexible side-chains at the interface; 4) semi-flexible SA with fully flexible interface (both backbone and side-chains). Practically, during the TAD-SA and the water refinement, the amino acids at the interface (side chains and backbone) are allowed to move to optimize the interface packing. The interface amino acids allowed to move are defined by the active and passive amino acids used in the AIRs ± 2 sequential amino acids.

The third docking step is a flexible explicit solvent refinement. In this final step, the structure obtained after the semi-flexible SA is refined in an explicit solvent layer (8Å shell of TIP3P water molecules). In this step, no real significant structural changes are expected; however, the scoring of the various structures is improved. The final structures are analysed and each is assigned a score, called "HADDOCK Score", which is a sum of several energy parameters: term intermolecular electrostatic (Elec), a term of van der Waals (vdW), a term of desolvation (Dsolv), a term energy of AIR and a term of buried surface area (BSA).

The "rigid-body score" (RBS) is equal to:

$$RBS = 1 * Elec + 1 * vdW - 0,05 * BSA + 1 * Dsolv + 1 * AIR$$

Equation 1.2. Rigid-body score HADDOCK

The final score is:

$$FS = 1 * Elec + 1 * vdW + 1 * Dsolv + 1 * AIR$$

Equation 1.3. Final score HADDOCK

Subsequently, the results are grouped in clusters, based on the RMSD (Root Mean Square Deviation) of each structure with respect to the reference structure (the structure to lower HADDOCK Score), using a cut-off determined manually by the operator (generally, around 7Å).

$$RMSD = \sqrt{\frac{1}{N} \sum_{i=1}^{i=N} \delta_i^2}$$

Equation 1.4. RMSD HADDOCK

In equation 1.4, the δ is the distance between the pair of equivalent atoms N (generally, are used C_α and sometimes C, N, O, C_β). In essence, the RMSD is the quadratic differences of the average distance of all C_α of a molecule compared with the same

equivalent C_{α} of another molecule and it measures the similarity of two 3D structures of two molecules following the superposition of the structures, whilst maintaining them rigid. To assess which model is best, one cannot rely on a single parameter, but some considerations must be made both on the energy, the frequency of a certain model (ie, how many times this model back in the dock, or how high the probability that is found in nature), and on the "visual appearance" of the complex generated. The HADDOCK (Van Zundert et al., 2016 and Dominguez et.al., 2003) program assigns to each structure and each cluster a score, HADDOCK score, which takes into account all the energy parameters. The complex is energetically stable as the HADDOCK score is lower. Obviously, the more stable energy does not always indicate the goodness of the structure. This because it can be that the real model in vivo is not necessarily the more energetically stable. For this reason, in addition to considering the scores energy, the user should consider how many times a certain "type" of structure is returned. In other words, the cluster is created, grouping all the structures that have a similar RMSD, which results in a structural similarity. Depending on how a cluster is populated, we can understand how likely a certain type of interaction takes place. Taking into account the abundance of a cluster against his HADDOCK score can be a good method in understanding whether a structure is reliable. Of course, it is necessary to look at the pdb file of the various models generated to understand if the program has created models consistent with the input data. It can happen that, to minimize the energy, the software chooses not to interact with an extended area of a molecule: obviously, such a structure, despite having a very good HADDOCK score, is definitely not a good model.

1.2.3 Autodock suite

The Autodock suite is a state of the art package that is used in numerous laboratories and for different application such as docking and virtual screening. Indeed, as stated in the Forli et.al., 2016 paper, Autodock can boast over 1,000 citations in the past year.

Autodock suite is free available and comprises several tools e.g. Autodock (Morris et al., 2009), Autodock Vina (Vina) (Trott and Olson, 2010), Raccon2 (Forli et.al., 2016), AutoDockTools (ADT) (<http://autodock.scripps.edu/resources/adt>), AutoLigand (Harris et.al., 2008). In this thesis, two computational docking programs (Autodock and Vina) have been used and tested.

Vina (Trott and Olson, 2010) uses a simple scoring function and rapid gradient-optimization conformational search while Autodock (Morris et al., 2009) relies on an empirical free-energy force field and rapid Lamarckian genetic algorithm search method. Vina (Trott and Olson, 2010) is a very fast method but uses several approximations in the scoring function. Indeed, the latter does not include any electrostatic terms, hydrogen atoms and furthermore it uses spherically symmetric hydrogen bond potentials to model hydrogen bond interactions. Vina has shown to give good results when ligands have typical biological size and composition (Trott and Olson, 2010).

On the other hand, the Autodock score functions comprises physically based terms. Indeed, the latter includes explicit polar hydrogen, a term for the electrostatic contribution and a term for the directional hydrogen-bonding interactions. The Autodock score function is already optimized to give successful results for different biological systems (Forli et.al., 2016) and it is made available to the user in the official web site

(<http://autodock.scripps.edu/resources/parameters>) allowing the expert users to calibrate the scoring function parameters depending on the particular system under investigation (Forli et.al., 2016).

Both programs use stochastic search method and predict a set of optimal docked poses, which are usually cluster spatially to verify the consistency of the results. However, it is worth mentioning that, by definition, stochastic methods do not guarantee that the global minimum is found.

Finally, to reduce the volume of the conformation space and thus the computational time, cost, both algorithms keep rigid the receptor molecule. As the matter of fact, the latter can be considered the most approximation in the Autodock/Vina based docking protocols.

Autodock docking steps. The Autodock docking (Morris et al., 2009) experiment includes at least seven steps, which are described as follows: 1) preparing the receptor molecule. In this step a .pdbqt file is generated, which contains information both hydrogen atoms and partial charges. By default Autodock uses “Gastaiger” method to calculate the partial charge. The Gastaiger” charge are calculated by using a fast empirical scheme. 2) preparing the ligand molecule. This step is similar to the previous step, but when the .pdbqt file is generated, the information of the ligand flexibility is included. 3) generating of the grid parameters file. Here the volume of the space (usually around the binding site of a receptor) that Autodock considers for the docking, is defined. This step generate the input file for “AutoGrid4” tool, which outputs several “map” and “grid” data file. 4) generating of the maps and grid data file. In this step, a set of maps and grid file are generated by AutoGrid4.2 script (Morris et al., 2009). 5) generating of the docking parameters file. Here, all the information for running Autodock are prepared. 6) running Autodock. 7) Results and analysis. Here, the complexes together with the estimate free energy of binding can be analyzed by using the graphical user interface ADT.

Vina docking steps: Vina (Trott and Olson, 2010) handles the same input file format i.e .pdbqt. Step1) and step2) remain unaltered, but Vina needs a particular file called “conf.txt” (see Fig.3.3), which contains all the docking parameters for the execution. Once the Vina job is launched, the program usually after few minutes returns the ligand poses (usually 10) with their estimate energy affinity.

```

Input:
--receptor arg      rigid part of the receptor (PDBQT)
--flex arg          flexible side chains, if any (PDBQT)
--ligand arg         ligand (PDBQT)

Search space (required):
--center_x arg      X coordinate of the center
--center_y arg      Y coordinate of the center
--center_z arg      Z coordinate of the center
--size_x arg         size in the X dimension (Angstroms)
--size_y arg         size in the Y dimension (Angstroms)
--size_z arg         size in the Z dimension (Angstroms)

Output (optional):
--out arg           output models (PDBQT), the default is chosen based on
                    the ligand file name
--log arg           optionally, write log file

Misc (optional):
--cpu arg           the number of CPUs to use (the default is to try to
                    detect the number of CPUs or, failing that, use 1)
--seed arg          explicit random seed
--exhaustiveness arg (=8) exhaustiveness of the global search (roughly
                    proportional to time): 1+
--num_modes arg (=9) maximum number of binding modes to generate
--energy_range arg (=3) maximum energy difference between the best binding
                    mode and the worst one displayed (kcal/mol)

Configuration file (optional):
--config arg        the above options can be put here

Information (optional):
--help              display usage summary
--help_advanced     display usage summary with advanced options
--version           display program version

```

Figure 1.3. Contents of a typical conf.txt file from Autodock tutorial. From Vina tutorial (<http://vina.scripps.edu/manual.html>).

1.2.4 LeapFrog (Sybyl)

LeapFrog algorithm (Tripos Inc., Tripos, St. Louis, MO, USA), a component of Sybyl (Tripos Inc) (v. 7.3) modelling suite, is used to screen virtual library of ligands/functional monomers (FMs) for their possible interactions with the receptor/templates. There are two different stages to calculate the energies: i) monomer-template complexation and ii) a system of scoring the complementarities between the monomer and template. In the latter, LeapFrog (Tripos Inc., Tripos, St. Louis, MO, USA) defines the template as the receptor binding site (using additional site-point matching scores, a system of scoring the receptor and ligand interactions).

This two-stage scoring is used to search a kind of regionality, therefore the “best fit” and the highest binding energy (in kcal/mol) will happen the best score is associated to complementary sites, which are nearby. The parameters in LeapFrog algorithm (Tripos Inc., Tripos, St. Louis, MO, USA) (see Fig.3.4) are standard and upon which the binding score is calculated. They have been optimised to favour hydrogen bonding capacity between small, finite molecules, and thus place less weight on other interactions such as hydrophobic, π – π or aromatic bonds.

Parameter name	Description	Default weight	Optimised weight
Join	Join different fragments	2	0
Fuse	Fuse candidate fragments	0	0
New	new ligand is started by aligning fragments	5	10
Fly	Alternative minimum energy ligand orientations	2	0
Twist	Conventional minimisation	2	5
Refine	Improves newly identified ligands	2	0
Bridge	Considers all fragments as bridges	2	0
Complement	Chooses moiety complementary to a cavity group as a ligand	2	0
Save	Saves ligands that satisfy specifications	2	5
Weed	Discards all but the 10 best ligands	0	1
Crossover	Generates best hybridizations among similar molecules	0	0
Prune	Can delete moieties of a known ligand on the basis of their energy with the receptor	0	0

Figure 1.4. Description of the relative move frequencies used by the Leapfrog algorithm to assign a docking mode and affinity score to the intermolecular complex.

1.3 Molecular Dynamics

In the previous sections, we focused only on static prediction techniques. In fact, we investigated how generate a 3D model of the protein from its sequence (homology modeling) and how generate a protein-protein complex or a protein-ligand complex (molecular docking). However, the biological processes occur in time. Thus, we need a method to predict the dynamical behavior of biomolecules. This method is called molecular dynamics (MD): a computer simulation approach where the time evolution of a set of interacting atoms is followed by integrating their equations of motion.

1.3.1 Choice of simulation approach: length and time scales

Molecular processes present time scale that span from the fs to the second and from pm to μm . In principle, we should simulate each biological system by using the Schrödinger equation. In this case, the computation time and cost would be infinite so, we must choose the correct computation approach in agreement with the time and lengths scale of the system that we want to investigate (see Fig.3.5). It is certain that we need to find a compromise between simulation speed and accuracy in the description of the system. Broadly speaking we can present three levels of detail:

1. Quantum mechanics (QM)

2. Atomistic molecular mechanics (MM)
3. Coarse-grained molecular mechanics (CG)

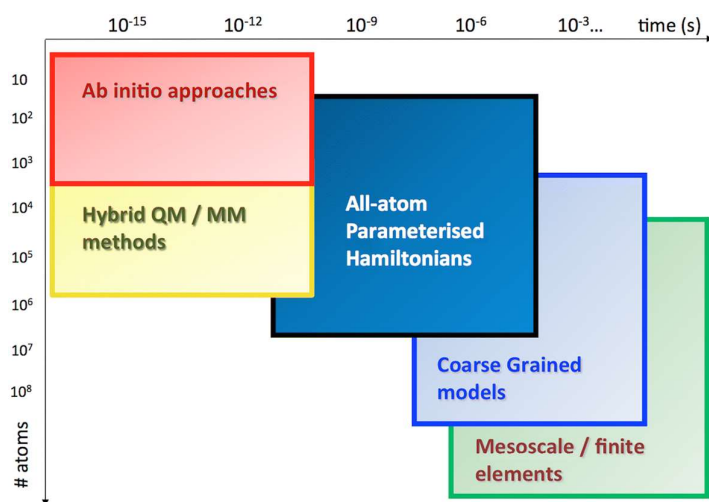


Figure 1.5. Accessible time- and size-scales for different computational methods in routine studies using state of the art implementations and architectures. Retrieved from Cascella and Vanni, 2015.

Originally, the methodology of MD was implemented before the 1960s and nowadays, many MD programs are available and the most famous are: GROMACS (Pronk et.al., 2013), AMBER (Salomon-Ferrer et.al., 2013), CHARMM (Brooks et.al., 2009) and NAMD (Phillips et.al., 2005). GROMACS (Pronk et al., 2013) was used in this thesis; however, the following generic aspects described in the following sections are valid for all MD software¹.

1.3.2 Molecular dynamics: introduction

Molecular mechanics (MM) methods enable to study timescales spanning from 10^{-12} to 10^{-5} s, and systems of ca. 10^6 atoms (Hug, 2013). Thus, systems e.g. proteins, in complex or not with ligand/protein in a solvent environment can be investigated, as well as the folding of short peptides. There are many MM techniques such as Monte Carlo (MC) simulations, energy minimization etc. However, all these approach does not offer any information about the biomolecule behaviour over the time i.e. the dynamics. MD can give information of the evolution of a molecular system in time. The smallest component of a system is the atom and these are considered as point in space which follow the Newton's 2nd law: where the time evolution of a set of interacting atoms is followed by integrating (at discrete timestep) their equations of motion. The force field defines the force by which the atom interacts. The force field is a mathematical description of both

¹ In this section, we will provide information about the MM and CG approaches because they were extensively used in this thesis. Although we are aware that QM methods are very important in the field, they are out of the aims of this thesis and thus are not described here.¹

bonded and nonbonded interactions, as depending on distances and angles between (two or more) atoms. This approximation implicitly includes all the electrons and their interactions but it does not provide any chemical reaction information. Again, MM is able to predict only conformational changes in the biological system, keeping the identity of molecules invariable along the simulation. In perfect conditions, the macroscopic (thermodynamic) calculable properties of the simulated ensemble can be reproduced by an extensive MD experiment. It does so under the ergodic hypothesis that if one allows the system to evolve on the limit of infinite time that system will eventually pass through all possible state.

Actually, the ergodicity is not achieved so, usually, it is a good compromise consider the convergence of a simulation when there are not alterations in the observables in time.

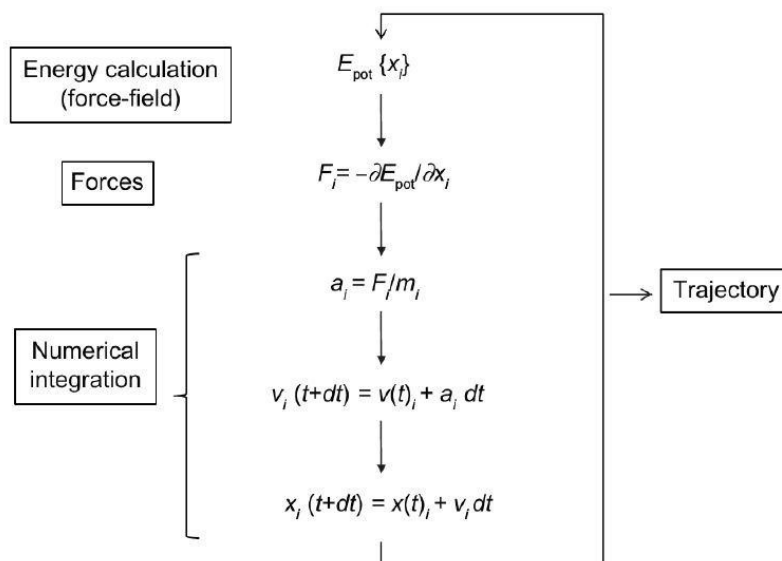


Figure 1.6. Workflow of molecular dynamics. The basic output is the trajectory, a file containing the positions (and possibly velocities and forces) of all the atoms in the system. Diagram retrieved from Sandal, 2016.

1.3.3 Periodic boundary conditions

The biological system under investigation is finite so boundaries must be considered. In principle, there are two options that can be adopted. In the first choice, the system can be inserted in a virtual box, thus it has a sort of “walls”. Consequently, this could lead to surface effects (artifacts) in the MD simulation because the behavior of atoms in proximity of the walls is unphysical by definition. Thus, the solution to this problem, more frequently used, is the periodic boundary conditions (PBC). On these terms, there are no rigid and impenetrable boundaries. Indeed, the simulation occurs in a finite but unbounded topology, that corresponds to the 3-dimensional surface of a 4-torus. Atoms that get out in on one side, they get in on the opposite side, seamlessly. In practical terms, the box results to be infinitely replicated in each direction (see Fig.3.7). PBC can still produce minor artifacts. Indeed, the user, during the MD setup, must carefully evaluate

the volume of the virtual box to avoid interactions between the molecule of interest and its own specular images (that is something of unphysical). The box shape (cubes, parallelepipeds, rhombic dodecahedrons) should be able to minimize the holes in the simulation space. Thus, spherical PBC box are not possible.

However, it is a good practice to simulate the system (usually in explicit solvent) in larger boxes than theoretical box size and the electrostatic treatment of PBC requires the box to be electrically neutral (Hug, 2013).

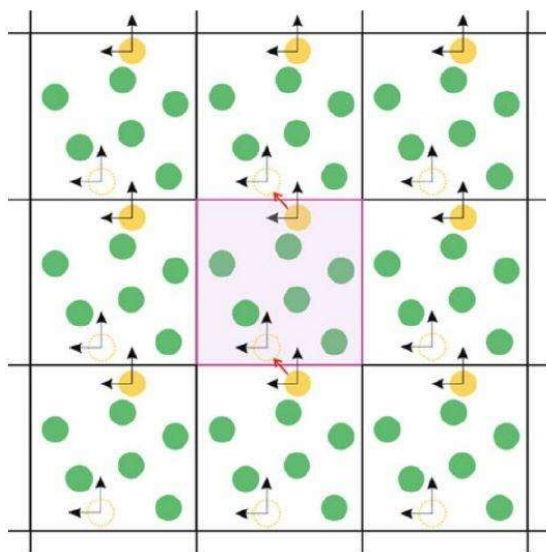


Figure 3.7. Schema describing the idea of PBC. The movement of the yellow particle (black arrows show the velocity components) is shown as moving from an edge of a box to the opposite edge. Retrieved from Hug, 2013.

1.3.4 Force Fields

A force field is a mathematical description of the force governing the interactions between atoms.

The interactions can be divided into two classes: i) bonded interactions and ii) non bonded interactions.

Bonded interactions include the covalent bonds between atoms, and model bond stretching, rotations, bending, dihedral angles etc. Bonded interactions can involve 2,3 or 4 atoms and can be described for example by harmonic potentials.

On the other hand, non bonded interactions include the electrostatic and van der Waals interactions, and are usually can involve 2 atoms. Van der Waals interactions are described by Lennard-Jones or Morse potentials. The properties of an atom strongly rely on its chemical environment. In view of this point, a force field characterizes the parameters for different atom types even for the same chemical element (Leach, 2001). Given a force field, all the information of the system such as atom types, atom connectivity are defined as “topology” of the system.

The most famous force fields are: AMBER (Salomon-Ferrer et.al.,2013), GROMOS (Oostenbrink et.al., 2004), CHARMM (Brooks et.al., 2009) and OPLS (Jorgensen et.al., 1996).

GROMOS force field was used in this thesis and it is a united atom force field, this means that non-polar hydrogens are not modelled explicitly. Force fields' parameters can be calculated by using ab initio methods or experimental data.

While GROMOS is parametrized with the purpose of reproduce the free enthalpies of hydration and solvation (Oostenbrink et.al.,2004), AMBER (Salomon-Ferrer et.al., 2013) values are generated by ab initio QM calculations (Cornell et.al., 1995).

It is worth mentioning that since different force fields are generated by using different strategies, they cannot mixed in the MD setting up.

1.3.5 Solvation

Biomolecule should be model considering the solvent effect. The solvent in MD simulation can be considered in two ways:

i) *Explicitly Solvent*. Explicit molecules of solvent (usually water) are included in the simulation. This means that more atoms must be simulated, augmenting the computational time. Actually, the solvent models (e.g. SPC/E (Berweiger et.al., 1995), TIP3P (Berendsen et.al., 1987)) still are not able to fully reproduce the water chemical-physical parameters. Indeed, in general water models show faster diffusion and lower internal structure than real water (Mark and Nilsson, 2001). Usually the water models with more than three atom points are used because they are able to better approximate the charge distribution of water (Mark, P. and Nilsson, L., 2001).

ii) *Implicitly Solvent*. In order to reduce the computational time and cost in MD simulations but at the same time keeping a certain level of accuracy, a new approach for solvent modeling has been developed. Implicit solvent methods model the solvent by a continuum potential and usually they treat the polar part as a continuous dielectric. The nonpolar component of solvation is estimated from the solvent-accessible surface (Chen et.al., 2008).

1.3.6 Cut-off

The calculation of nonbonded terms in the potential energy functions represents the most consuming part of MD simulations. The values of these terms grow in a quadratic manner with the number of atoms present in the system. For instance, in a system composed by 100 atoms, the number of interactions to be considered are about 5000, if we have 10000 atoms, the interaction became about 50×10^6 . In these cases, to make more efficient the calculation, the interactions between atoms that are at a distance greater than a certain threshold (cut-off) can be neglected. Of course, ignoring completely the interactions that take place at distance greater ($>$ cut-off) is a simplistic approach and can generate artifacts in the simulation. To overcome the problem several approaches have been developed. The most common approaches are three: i) truncation method, ii) shift cut off method and iii) switch cut off method.

In the truncation method for the interatomic distances greater than the cutoff, the interactions are simply set to zero. The shift cutoff approach modifies the entire potential energy surface such that at the cutoff distance the interaction potential is zero. The switch cutoff approach reduces the interaction potential over a predefined range of distances.

The potential takes its usual value up to the first cutoff and is then switched to zero between the first and last cutoff
(http://www.ch.embnet.org/MD_tutorial/pages/MD.Part3.html).

1.3.7 Integration of the equation motion

Given a system of N particles, and a force field:

$$U(\vec{r}_i), i = 1, \dots, N$$

Equation 1.5. Force Field.

we have a $3N$ second-order differential equation set which represent Newton's equations of motion:

$$\vec{F}_i = m_i \ddot{\vec{r}}_i$$

Equation 1.6. Newton's second law.

In MD, we discretize differential equations as difference equations, integrated over discrete time steps. Anyhow, our goal is to simulate something where the time should be continuous. Thus, an important factor in MD simulation is the timestep. A large timestep would coarse grain the integration too much and produce unphysical artefacts. As the matter of fact, if we decided to simulate the system with a small timestep, the computation time will be high. The choice thus usually falls on the largest possible timestep that falls below half the characteristic time of the fastest motions of the system.

The fastest movements are the vibrational movements (bond stretching and bending), which restrict our time step to 1-4 fs.

The integrator algorithm, in agreement to Newton's equations, should be also in theory time-reversible and symplectic (Hug, 2013 and Frenkel and Smit, 2002). An integrator algorithm is defined "symplecticness", when the phase space volume and total energy are preserved in time and thus, it must follow the Liouville's theorem. The most famous symplectic algorithms are: Verlet (Verlet, 1967), Velocity Verlet (Verlet, 1968 and Swope et al., 1982) and Leapfrog (Hockney and Eastwood, 1988).

1.3.8 Thermostats

Given that Newton's equations conserve energy, the integration of Newton's equations of motions would direct the simulation to belong to a microcanonical (NVE) ensemble (Frenkel and Smit, 2002). However, we need to recreate situations that are close to the biological reality, where the temperature is preserved (NVT or NPT ensembles). To perform this, we need to mimic the coupling of our system to a heat bath. The latter is

carried out by a thermostat algorithm, which correct the velocities of particles in order to recreate, with different degrees of approximation, a canonical ensemble.

The simplest method would consist rescaling the velocities so that they conserve, at each step, the Boltzmann distribution at the fixed temperature. Velocity rescaling, however, would damp or erase the thermal fluctuations of the system, producing an artificial ensemble. Many other algorithms exist, anyway, a detailed description of these is out of the scope of the present thesis.

1.3.9 Coarse grained methods

Many approaches have been proposed in order to save computational time and cost for MD simulations. One of these methods is called “Coarse grained (CG)”. The latter by coarse graining the molecular topology, reduces the multiple real atoms to a single coarse-grained component by a mapping process (see Fig.3.8) (Cascella and Vanni, 2015). In CG simulations, most of the MM basis and algorithms are valid. Indeed, GROMACS (Pronk et.al., 2013), the software for MD simulations, can work both with MM and CG atom representation; and the difference between MM and CG is in the force field used.

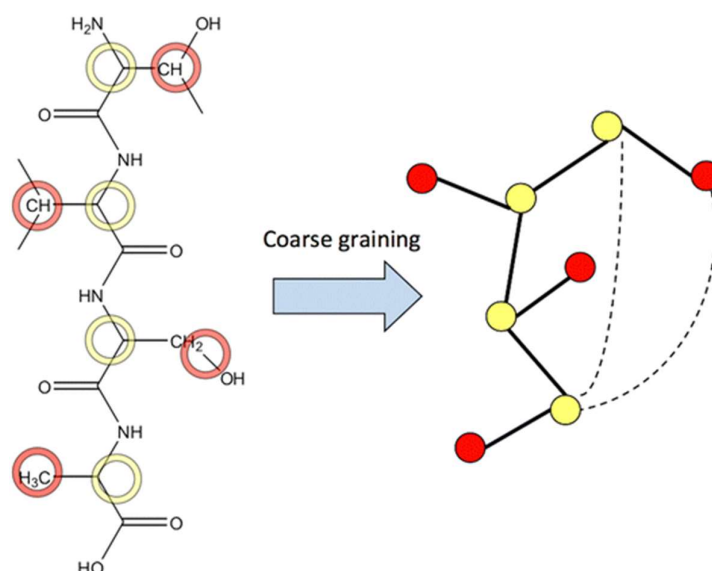


Figure 1.8. Example of mapping between an atomistic polypeptide chain (left) and a CG representation (right). Retrieved from Cascella and Vanni, 2015.

CG force fields and mappings are often generated using the same basis (reproducing experimental data sets, chemical-physical quantities), however different mappings and force fields (with different level of details) can be adopted based on what we want study (Cascella and Vanni, 2015). The two CG methodologies that have shown successful results are: MARTINI and Go models. In this thesis, MARTINI force field has been used and we will provide a brief description below.

1.3.9.1 MARTINI

MARTINI is both a CG force field and mapping protocol that, as reported in literature, has been exploited mostly for protein-lipid membrane MD simulation (Monticelli et.al., 2008 and Ceccon et al., 2015). The MARTINI parametrization follows the same idea of a MM force field and it is similar to GROMOS (Christen et.al., 2005) force field.

Non bonded force field parameters are described by the Lennard-Jones potentials. The latter attempt to reproduce the general chemico-physical quantities (called, oil/water partition coefficients) rather than a particular structural conformation.

On the other hand, bonded parameters rely on statistical polypeptide properties and are retrieved from PDB database. Generally, the mapping strategy is 4-to-1, i.e. 4 MM atoms correspond to 1 CG atom and the electrostatic is considered explicitly (Monticelli et.al., 2008). MARTINI includes the biomolecule structure information, since the secondary structure has to be set up at the start and restrained.

